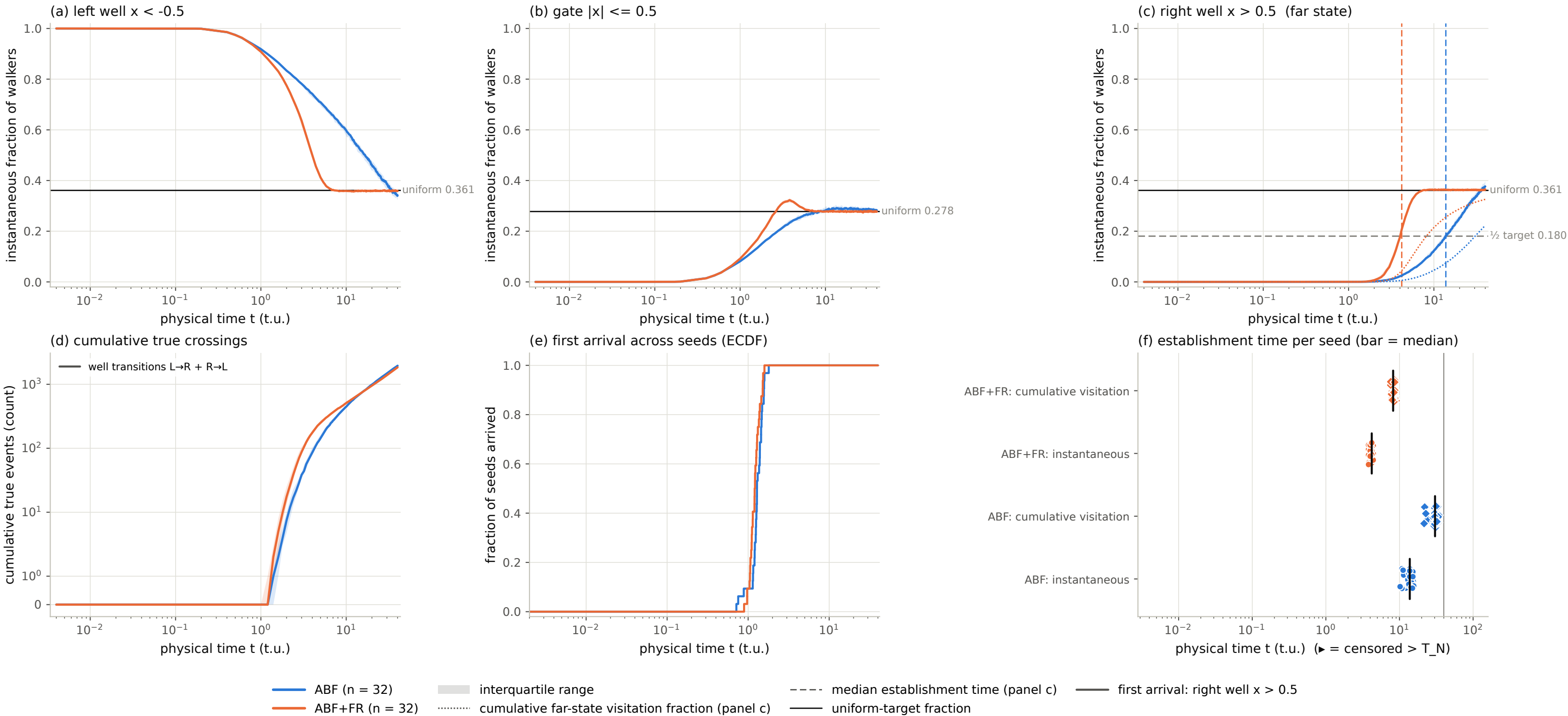


Establishment and crossings vs physical time



Line = median across seeds; band = interquartile range (25th-75th percentile) across seeds. Metrics are the frozen SCIENTIFIC_PLAN section-4 quantities, unsmoothed.

Establishment (frozen): first save after which the instantaneous far-state fraction (right well $x > 0.5$) stays $\geq \frac{1}{2}$ of its uniform target 0.361; the cumulative-visitation version uses C_all (far-state bins) against $\frac{1}{2}$ of their bin share. True events never count a copy (clones inherit labels).

Null calibration at N = 2048: an EXACTLY uniform population (far count $\sim \text{Bin}(N, 0.361)$ per save) fails the instantaneous criterion at a save with $p = 1.74e-72$ and passes every save of the second half with $P = 1.000$.